ABBREVIATIONS

A	AMPERES
AC	ALTERNATING CURRENT
ACT	ACOUSTIC TILE
ADJ	ADJUSTABLE
AF	AMPERE FRAME
AFF	ABOVE FINISH FLOOR
AIC	AMP INTERRUPTING CAPACITY
AL	ALUMINUM
ALT	ALTERNATE
AM	AMMETER
ARCH	ARCHITECTURAL
ASC	ABOVE SUSPENDED CEILING
ASD	ADJUSTABLE SPEED DRIVE
AT	AMPERE TRIP
ATS	AUTOMATIC TRANSFER SWITCH
AUTO	AUTOMATIC
AVG	AVERAGE
AWG	AMERICAN WIRE GAUGE
BAS	BUILDING AUTOMATION SYSTEM
BCW	BARE COPPER WIRE
BD	BUS DUCT
BDF	BUILDING DISTRIBUTION FRAME
BFC	BELOW FINISH CEILING
BFF	BELOW FINISH FLOOR
BHP	BRAKE HORSE POWER
BOB	BOTTOM OF OUTLET BOX
BLDG	BUILDING
BS	BEST BATTERY SELECTOR (DC POWER)
BTP	BREAKER TEST PANEL
BKR	BREAKER
C	CONDUIT
CAB	CABINET / CAM LOCK CONNECTION
CCT	CIRCUIT
CCT BKR (CB)	CIRCUIT BREAKER
CCTV	CLOSED CIRCUIT TELEVISION
CIRC	CIRCULATING/CIRCULATE
CFCI	CONTRACTOR FURNISHED CONTRACTOR INSTALLED
CKT	CIRCUIT
CL	CENTERLINE
CLG	CEILING
CMU	CONCRETE MASONRY UNIT
CO	CONDUIT ONLY
COM	COMMUNICATION
COMB	COMBINATION
CONN	CONNECTION
CONTR	CONTRACTOR
CORR	CORRIDOR
CP	CONTROL PANEL
CPT	CONTROL POWER TRANSFORMER
CS	CONTROL SWITCH
CT	CURRENT TRANSFORMER
CTR	CENTER
CU	COPPER
CUH	CABINET UNIT HEATER
D	DEDICATED
DC	DIRECT CURRENT
DCP	DC POWER DISTRIBUTION PANEL
DCTS	DC POWER TIE SWITCH
DET	DETAIL/DETECTOR
DIA	DIAMETER
DISC	DISCONNECT
DN	DOWN
DP	DISTRIBUTION PANEL
DPDT	DOUBLE POLE DOUBLE THROW
DS	DISCONNECT SWITCH
DWG	DRAWING
EA	EACH
EC	ELECTRICAL CONTRACTOR
EF	EXHAUST FAN
EG	EQUIPMENT GROUND
EGS	ENGINE GENERATOR SET
EHT	ELECTRIC HEAT TRACING
EJ	EXPANSION JOINT
EJB	ELECTRICAL JUNCTION BOX
EL	ELEVATION
ELEC	ELECTRIC/ELECTRICAL
ELP	EMERGENCY LIGHTING PANEL
ELC	EMERCENCY LIGHTING CONTACTOR
EMER	EMERGENCY
EMCC	EMERGENCY MOTOR CONTROL CENTER
EMH	ELECTRICAL MANHOLE
EMT	ELECTRICAL METALLIC TUBING
EQUIP	EQUIPMENT
ERP	EMERGENCY RECEPTACLE PANEL
ES	EMERGENCY STOP OR SWITCH
ETR	EXISTING TO REMAIN
EWC	ELECTRIC WATER COOLER
EXIST (E)	EXISTING
EXT	EXTERIOR
(F)	FUTURE
F	FREQUENCY METER/FUSE
FA	FIRE ALARM
FACP	FIRE ALARM CONTROL CABINET
FATC	FIRE ALARM TERMINAL CABINET
FB	FLOOR BOX
FBO	FURNISHED BY OTHERS
FDR	FEEDER
FDH	FACILITIES DATA HISTORIAN
FH	FUME HOOD
FIN	FINISH
FIXT	FIXTURE
FLA	FULL LOAD AMPERES
FLEX	FLEXIBLE
FLR	FLOOR
FLUOR	FLUORESCENT
FMS	FACILITIES MONITORING SYSTEM
FOP	FIBER OPTIC PATCH PANEL
FP	FIREPROOF/FIRE PROTECTION
FRPT	FIRE RATED POKE THROUGH
FS	FLOW SWITCH/FIRE-SMOKE
FSD	COMBINATION FIRE & SMOKE DAMPER
FSS	FUSED SAFETY SWITCH
FVNR	FULL VOLTAGE NON REVERSING
FWE	FURNISHED WITH EQUIPMENT
G	GROUND
GA	GAUGE
GALV	GALVANIZED
GC	GENERAL CONTRACTOR
GD	GROUND DETECTOR
GEN	GENERATOR
GFCI	GROUND FAULT CIRCUIT INTERRUPTER
GFI	GROUND FAULT INTERRUPTER
GND	GROUND
GMD	GROUND MONITORING DEVICE
GRC	GALVANIZED RIGID CONDUIT
GTB	GROUND TERMINAL BOX
H	HAND
HH	HANDHOLE
HJB	HEAT TRACING JUNCTION BOX
HOA	HAND-OFF-AUTO
HMI	HUMAN MACHINE INTERFACE
HP	HORSEPOWER
HRG	HIGH RESISTANCE GROUNDING
HT	HEIGHT/HEAT TRACE
HTR	HEATER
HV	HIGH VOLTAGE
IDF	INTERMEDIATE DISTRIBUTION FRAME
IFS	INTEGRATED FACILITY SYSTEM
I/I	CURRENT TRANSDUCER
IMC	INTERMEDIATE METAL CONDUIT
INCAND	INCANDESCENT
INST	INSTRUMENT
INT	INTERIOR
INTERC	INTERCONNECTION
IPC	INTEGRATED POWER CENTER
JB	JUNCTION BOX
JOA	JOG-OFF-AUTO
KCMIL (MCM)	THOUSAND CIRCULAR MILS
KV	KILOVOLT
KVA	KILOVOLT-AMPERES
KW	KILOWATTS
KWH	KILOWATT HOURS
L	LENGTH
LAN	LOCAL AREA NETWORK
LB	POUND
LBS	POUNDS
LCP	LOCAL CONTACTOR PANEL
LCS	LOCAL CONTROL STATION
LP	LIGHTING PANEL
LTG	LIGHTING
LTX	LIGHTING TRANSFORMER
LV	LOW VOLTAGE
MAN	MANUAL
MAX	MAXIMUM
MC	MOTOR CONTROLLER
MCB	MAIN CIRCUIT BREAKER
MCC	MOTOR CONTROL CENTER
MCP	MOTOR CIRCUIT PANEL
MCM (KCMIL)	THOUSAND CIRCULAR MILLS
MDP	MAIN DISTRIBUTION PANEL
MECH	MECHANICAL
MER	MECHANICAL EQUIPMENT ROOM
MFR	MANUFACTURER
MH	MANHOLE
MIN	MINIMUM
MLO	MAIN LUGS ONLY
MPOE	MAIN POINT OF ENTRY
MRCT	MEGGER RELAY CURRENT TRANSFORMER
MS	MANUAL STARTER
MTCP	MOTOR CIRCUIT PROTECTOR
MTD	MOUNTED
MTG	MOUNTING
MTG HT	MOUNTING HEIGHT
MTR	METER/MOTOR
(N)	NEW
NA	NOT APPLICABLE
NC	NORMALLY CLOSED
NFD	NON-FUSED DISCONNECT
NFSS	NON-FUSED SAFETY SWITCH
NIC	NOT IN CONTRACT
NO	NORMALLY OPEN
NGR	NEUTRAL GROUNDING RESISTOR
NOM	NOMINAL
NTS	NOT TO SCALE
OC	ON CENTER
OD	OUTSIDE DIAMETER
OFCI	OWNER FURNISHED, CONTRACTOR INSTALLED
OFOI	OWNER FURNISHED, OWNER INSTALLED
OHM	OHMMETER
P	PENDANT/POLE/PUMP
PA	PUBLIC ADDRESS
PB	PULL BOX
PC	PHOTOCELL
PDU	POWER DISTRIBUTION UNIT
PFM	POWER FACTOR METER
PH	PHASE
PI	POSITION INDICATOR
PLBG	PLUMBING
PNL	PANEL
POC	POINT OF CONNECTION
POE	POWER OVER ETHERNET
PRES	PRESSURE
PRV	PRESSURE REDUCING VALVE
PT	POTENTIAL TRANSFORMER
PWR	POWER
P&ID	PROCESS AND INSTRUMENTATION DIAGRAM
REC	RECESSED/RECORDING
RECEPT	RECEPTACLE
(R)	RELOCATE
REQ	REQUIRED
REV	REVISION OR REVERSE
RGIP	REMOTE GROUND INDICATOR PANEL
RGM	REMOTE GENERATOR MONITORING PANEL
RJB	RTD JUNCTION BOX
RM	ROOM
RPM	REVOLUTIONS PER MINUTE
ROP	REMOTE OPERATING PANEL
RTD	RESISTIVE TEMPERATURE DETECTOR
SB	SPLICE BOX
SCHD	SCHEDULE
SD	SMOKE DETECTOR
SECT	SECTION
SH	SPACE HEATER
SHT	SHEET
SIG	SIGNAL
SIM	SIMILAR
SPEC	SPECIFICATION
SPD	SURGE PROTECTION DEVICE
SPST	SINGLE POLE SINGLE THROW
SPDT	SINGLE POLE DOUBLE THROW
SQ	SQUARE
SQ FT	SQUARE FOOT
SMR	SURFACE MOUNTED RACEWAY
SRCT	SPLIT CORE CURRENT TRANSFORMER
SS	STAINLESS STEEL
STA	STATION
STD	STANDARD
STL	STEEL
STR	STARTER
STRU	STRUCTURAL/STRUCTURE
SW	SWITCH
SWBD	SWITCHBOARD
SWG	SWITCHGEAR
TB	TERMINAL BLOCK
TC	TERMINAL CABINET
TEL	TELEPHONE
TEMP	TEMPERATURE
TOB	TOP OF RECEPTACLE BOX
THK	THICK
TS	TAMPER SWITCH/TIME SWITCH
TV	TELEVISION
TVSS	TRANSIENT VOLTAGE SURGE SUPPRESSION
TYP	TYPICAL
UPS	UNINTERRUPTIBLE POWER SUPPLY
UC	UNDER CABINET
U/G	UNDERGROUND
UH	UNIT HEATER
UNO	UNLESS NOTED OTHERWISE
US	UNIT SUBSTATION
V	VOLTS
VA	VOLT AMPERES
VAC	VOLT ALTERNATING CURRENT
VAR	VOLTS AMPERES REACTIVE
VARM	VARMETER
VD	VOLTAGE DROP
VDC	VOLTS DIRECT CURRENT
VH	VARHOUR METER
VM	VOLT METER
VP	VAPORPROOF
VS	VOLTMETER SWITCH
VSD	VARIABLE SPEED DRIVE
V/V	VOLTAGE TRANSDUCER
W	WIDTH/WIRE or WATT
W/	WITH
WH	WATTHOUR METER
WHD	W/DEMAND ATTACHMENT
WM	WATTMETER
WP	WEATHERPROOF
WR	WELDING RECEPTACLE
WS	WALL SURFACE
WT	WATER TIGHT
W-V	WATT-VAR INDICATING METER
WVS	WATT-VAR SELECTOR SWITCH
W/W	WATT TRANSDUCER
XFMR	TRANSFORMER
XP	EXPLOSION PROOF

* REFER TO PIPING, EQUIPMENT, AND INSTRUMENTATION SYMBOLS FOR ADDITIONAL ABBREVIATIONS.

GROUNDING

SYMBOL	DESCRIPTION
G # 4/0 AWG	BARE GROUNDING CONDUCTOR - BURIED, OR A ROUTED ON THE SIDE OF OF A CABLE TRAY , SIZE INDICATED ON DWG.
G # 4/0 AWG	GREEN INSULATED GROUNDING CONDUCTOR EXPOSED OR CONCEALED, SIZE INDICATED DWG.
G	BARE OR GREEN INSULATED GROUNDING CONDUCTOR TAP SEE PLAN DRAWING FOR CABLE TYPE AND SIZE
DIG	GREEN/YELLOW INSULATED GROUNDING CONDUCTOR BELOW FLOOR SLAB OR BURIED (ISOLATED GROUND FOR DELTA V ROOMS)
DIG	GREEN/YELLOW INSULATED GROUNDING CONDUCTOR - EXPOSED (ISOLATED GROUND FOR DELTA V ROOMS)
⊗	GROUND ROD 3/4" x 10' UNO
◎	GROUND ROD 3/4" X 10' UNO, WITH TEST WELL
G G G	GROUND ROD TRIAD, 3/4" X 10'-0" UNO (PLANT GROUND) WITH 4/0 AWG BONDING JUMPERS UNO
DIG DIG DIG	GROUND ROD TRIAD, 3/4" X 10'-0", UNO (DELTA V INSTRUMENT GROUNDING ELECTRODE) WITH 4/0 AWG BONDING JUMPERS UNO
U	CONCRETE ENCASED ELECTRODE (UFER) GROUNDED COLUMN
■	EXOTHERMIC WELD
●	BURNDY HYGROUND CONNECTION (CABLE TO CABLE)
+	COMPRESSION CROSS CONNECTION (CABLE TO CABLE)
▲	BURNDY HYGROUND GROUNDLINK CONNECTOR (YGIB) WITH TERMINAL LUG OR APPROVED EQUAL
▣	BOLTED GROUND CONNECTION TO EQUIPMENT OR MISC. STEEL WITH TERMINAL LUG OR SERVIT POST
▭ (MGB OR DIG)	MGB - MAIN GROUND BAR DIG - DELTA V ISOLATED GROUND BAR
—○	CONDUCTOR TURNS UP TOWARDS VIEWER (PENETRATES UPPER LEVEL)
—●	CONDUCTOR TURNS DOWN AWAY FROM VIEWER (PENETRATES LOWER LEVEL)
◆	AIR TERMINAL - LIGHTNING PROTECTION
◪	LIGHTNING PROTECTION THRU ROOF ASSEMBLY WITH FLASHING AND PITCH POCKET

LIGHTING

SYMBOL	DESCRIPTION	SYMBOL	DESCRIPTION
☰	2'x4' DIRECT/INDIRECT FIXTURE SEE FIXTURE SCHEDULES FOR CATALOG NUMBERS (TYP.)	○ OR □	CEILING RECESSED MOUNTED FIXTURE (ROUND OR SQUARE)
☰	2'x2' DIRECT/INDIRECT FIXTURE		RECESSED MOUNTED WALL WASH
☰	1'x4' DIRECT/INDIRECT FIXTURE		WALL SCONCE
☰	1'x2' DIRECT/INDIRECT FIXTURE	● ▬	"SHADING" INDICATES FIXTURES SUPPLIED FROM EMERGENCY SOURCE
□	2'x2' CLEANSED FIXTURE	⊗	CEILING MOUNTED EXIT SIGN SINGLE FACE WITH CHEVRON DIRECTIONAL
▭	2'x4' CLEANSED FIXTURE	⊗	CEILING MOUNTED EXIT SIGN DOUBLE FACE WITH CHEVRON DIRECTIONAL
▭	1'x4' CLEANSED FIXTURE	⊗	WALL MOUNTED EXIT SIGN SINGLE FACE
├○┤	PENDANT OR SUSPENDED (2', 3', 4', 8', 12', 16' LENGTHS) SEE LIGHTING FIXTURE SCHEDULE FOR LENGTHS AND LAMP SIZES	⊗	WALL MOUNTED EXIT SIGN DOUBLE FACE
LED	STRIP LED		STANCHION MOUNTED FIXTURE (SHADED DOT INDICATES CONDUIT FITTING)
C	LIGHTING CONTACTOR LOCATION: WALL HEIGHT: AS NOTED		STANCHION MOUNTED FIXTURE SUPPLIED FROM EMERGENCY SOURCE (SHADED SMALL DOT INDICATES CONDUIT FITTING)
D	ROOM CONTROL SWITCH HEIGHT: 3'-6" BOB TO FF., UNO		CEILING SURFACE MTD. FIXTURE
PC	PHOTO CELL LOCATION: WALL HEIGHT: AS NOTED		CEILING SURFACE MOUNTED FIXTURE SUPPLIED FROM EMERGENCY SOURCE
EP	EXTERIOR PHOTO CELL LOCATION: WALL HEIGHT: AS NOTED		PENDANT MOUNTED FIXTURE
DS	DAYLIGHT SENSOR LOCATION: CEILING		PENDANT MOUNTED FIXTURE SUPPLIED FROM EMERGENCY SOURCE
RC	ROOM CONTROLLER LOCATION:CEILING		WALL MOUNTED FIXTURE
PL	PLUG LOAD CONTROLLER LOCATION:CEILING		WALL MTD. FIXTURE SUPPLIED FROM EMERGENCY SOURCE
Ⓢ	OCCUPANCY SENSOR LOCATION: CEILING		WALL MOUNTED WALLPACK
Ⓢ	OCCUPANCY SENSOR FOR LIGHTING CONTROL LOCATION: WALL HEIGHT: 3'-6" FF, UNO		STEP LIGHT FIXTURE
		⊞	BOLLARD FIXTURE
			FLOOD LIGHT WITH AIM POINT INDICATION
		ab Ⓜ	OCCUPANCY SENSOR FOR DUAL LIGHTING CONTROL LOCATION: WALL HEIGHT: 3'-6" FF, UNO

LIGHTING SWITCHES

SYMBOL	DESCRIPTION	SYMBOL	DESCRIPTION
$ 6/xxxx	SINGLE POLE CKT NO./ROOM NO.	$D 6a	DIMMER CKT NO. AND SWITCH LEG
$ 6a	SINGLE POLE CKT NO. SWITCH LEG	$MS 6a	MOTORIZED SHADES SWITCH CKT. NO. AND SWITCH LEG
$2P 6a,7a	2 - POLE CKT. NOS. AND SWITCH LEGS	$SCN 6a	MOTORIZED SCREEN SWITCH CKT. NO. AND SWITCH LEG
$3 6L OR 6a	3 - WAY CKT. NO. (SUPPLY) OR SWITCH LEG	$TL 6a	TANK LIGHT CKT. NO. AND SWITCH LEG
$4 6a	4 - WAY CKT. NO. SWITCH LEG	$LV 6a	LOW VOLTAGE SWITCH CKT. NO. AND SWITCH LEG

SWITCH NOTES:

A. WALL-MOUNTED LIGHTING SWITCHES SHALL BE MOUNTED 4 FT. FROM CONCRETE FLOOR TO CENTER OF BOX, UNO.

B. CKT. NOS. SHOWN ARE EXAMPLES ONLY, SEE SWITCH LEG AND TRAVELLER NUMBERING SCHEME AT LEFT.

RECEPTACLES

SYMBOL	DESCRIPTION	SYMBOL	DESCRIPTION
	SIMPLEX RECEPTACLE MOUNTING HEIGHT: 1'-6" CTR OF BOX TO FF, UNO		DUPLEX RECEPTACLE FLOOR MOUNTED
	DUPLEX RECEPTACLE MOUNTING HEIGHT: 1'-6" CTR OF BOX TO FF, UNO		DOUBLE DUPLEX RECEPTACLE FLOOR MOUNTED
	QUADRUPLEX RECEPTACLE MOUNTING HEIGHT: 1'-6" CTR OF BOX TO FF, UNO		DUPLEX RECEPTACLE CEILING MOUNTED
	DUPLEX RECEPTACLES MOUNTED ABOVE FURNITURE HEIGHT NOTED ON DWGS.		DOUBLE DUPLEX RECEPTACLE CEILING MOUNTED
	QUADRUPLEX RECEPTACLES MOUNTED ABOVE FURNITURE HEIGHT NOTED ON DWGS.		RECEPTACLE ON STANDBY POWER (TYPICAL FOR ALL OTHER TYPES WITH CENTER SHADING CENTER SHADING MAY BE OMITTED IF ENTIRE BUILDING IS ON STANDBY POWER)
	DUPLEX RECEPTACLE PLUG LOAD CONTROLLED MOUNTING HEIGHT: 1'-6" CTR OF BOX TO FF, UNO	■	SYSTEM FURNITURE POWER AND TELCO POLE INCLUDING PLUG LOAD CONTROLLED AND UNCONTROLLED RECEPTACLES. PROVIDED AS PART OF FURNITURE SYSTEM
	QUAD RECEPTACLE WITH (2) NORMAL PLUGS AND (2) PLUGS LOAD CONTROLLED MOUNTING HEIGHT: 1'-6" CTR OF BOX TO FF UNO	◧	SYSTEM FURNITURE POWER POLE INCLUDING PLUG LOAD CONTROLLED AND UNCONTROLLED RECEPTACLES
	RECEPTACLE PEDESTAL MOUNTED	Ⓡ	OVERHEAD CABLE REEL WITH DUPLEX RECEPTACLE
	SURFACE MOUNTED WIREWAY, WITH RECEPTACLES (I.E., WIREMOLD)	480V 30A XX	SPECIAL PURPOSE RECEPTACLE VOLTAGE AND AMPERAGE AS INDICATED RECEPTACLE SUPERSCRIPT NOTES: D = DEDICATED E = EMERGENCY (RED) G = GFCI U = UPS (BLUE) CR = CLEAN ROOM S = SURFACE WR = WELDING RECEPTACLE WP = WEATHER PROOF XP = EXPLOSION PROOF NOTE: 120 V RECEPTACLES ARE NEMA 5-20R, UNO ON PLAN OR DETAIL
	4-GANG, FLOOR MOUNTED QUAD RECEPTACLE DUAL DUPLEX (NORMAL/UPS) PWR WITH DATA AND AV CONNECTIONS	F 60A 480V WR	WELDING RECEPTACLE, 60 A, 480 V, WITH FUSED DISCONNECT SWITCH
	4-GANG, FLOOR MOUNTED QUAD RECEPTACLE DUAL DUPLEX (NORMAL/UPS) PWR WITH DATA CONNECTIONS	B 60A 480V WR	WELDING RECEPTACLE, 60 A, 480 V, WITH CIRCUIT BREAKER
	2-GANG, FLOOR MOUNTED DUPLEX RECEPTACLE WITH DATA CONNECTIONS	ERXXXA BBBBBBBBB-CC	RECEPT OR SWITCH DETAIL NO. PANELBOARD AND CCT BKR (OUTLET ELEVATION NOTED ON DETAIL)
	6" DIA. FRPT FLR BOX, DOUBLE DUPLEX RECPT., NORMAL AND UPS POWER WITH DATA CONNECTIONS		
	4" DIA. FRPT FLR BOX, DUPLEX RECPT., NORMAL POWER, WITH DATA CONNECTIONS		

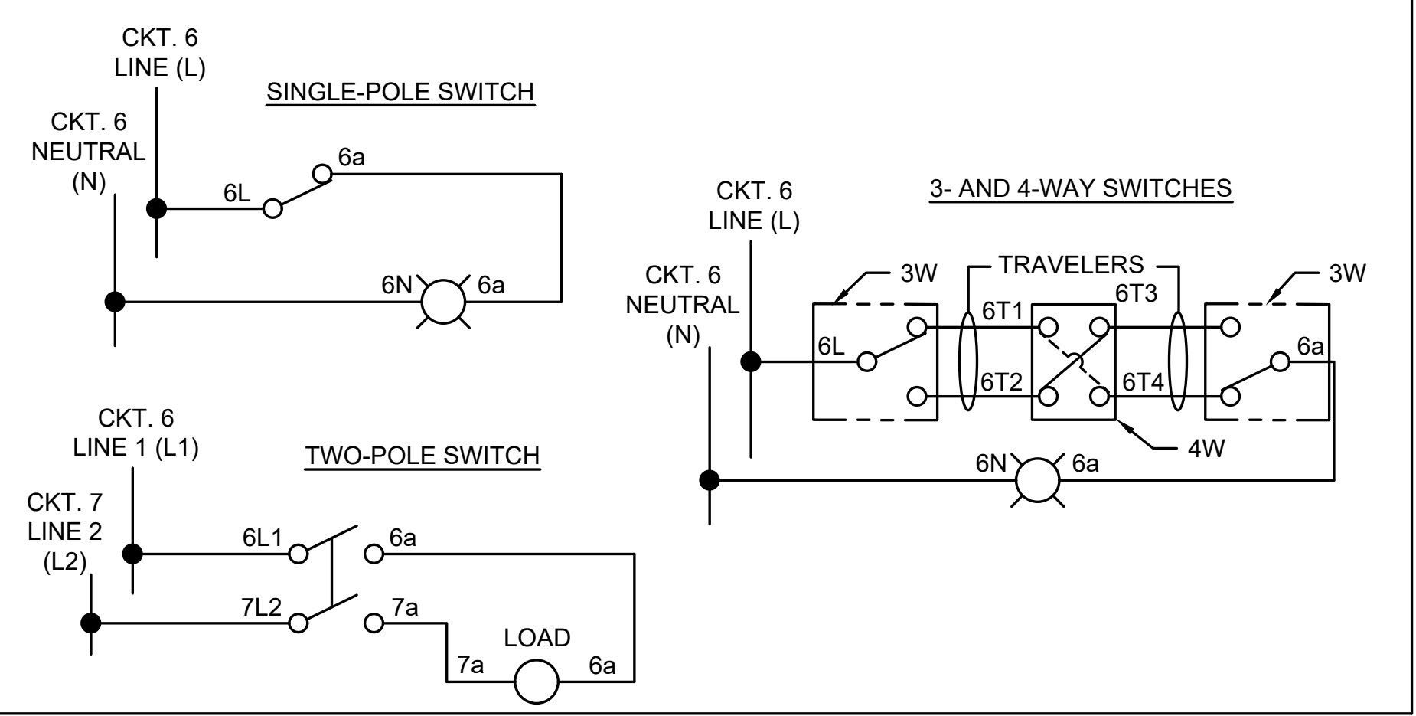
CONTROL STATIONS

SYMBOL	DESCRIPTION
○	STOP PUSHBUTTON
SS RD	START/STOP CONTROL STATION WITH RED PILOT LT SS = START/STOP RD = RED
SS	PUSH BUTTONS SS = START/STOP
RD	PILOT LIGHT WITH COLOR INDICATION A = AMBER BL = BLUE GR = GREEN RD = RED WH = WHITE
HOA	SELECTOR SWITCH ONLY HOA = HAND-OFF-AUTO JOA = JOG-OFF-AUTO O/F = ON-OFF
HOA RD	SELECTOR SWITCH AND PILOT LIGHT WITH COLOR INDICATION HOA = HAND-OFF-AUTO JOA = JOG-OFF-AUTO O/F = ON-OFF A = AMBER BL = BLUE GR = GREEN RD = RED WH = WHITE

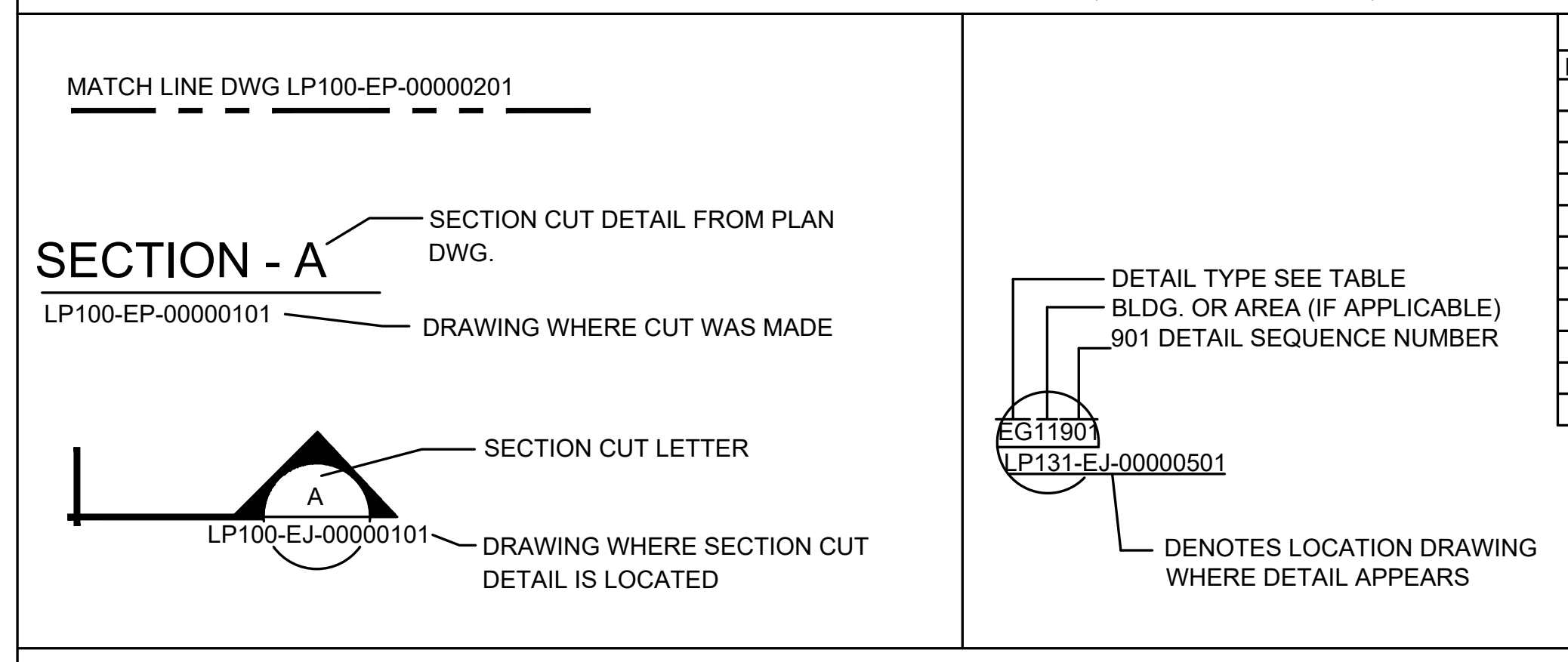
POWER

SYMBOL	DESCRIPTION
Ⓜ	MOTOR
▬	FLUSH MOUNTED PANEL
▬	SURFACE MOUNTED PANEL
▭	SWITCHGEAR/ SWITCHBOARD
⊠	CONTROL PANEL
☐☐☐	MOTOR CONTROL CENTER, VERTICAL SECTIONS INDICATED (3 SHOWN FOR EXAMPLE)
Ⓙ	JUNCTION BOX, WALL MOUNT 4-11/16" SQ. X 2-1/8" UNO
Ⓙ	JUNCTION BOX, CEILING MOUNT 4-11/16" SQ. X 2-1/8" UNO
JB	JUNCTION BOX, SIZED BY INSTALLATION CONTRACTOR
PB	PULL BOX, SIZED BY INSTALLATION CONTRACTOR
T 30 KVA	TRANSFORMER WITH KVA INDICATED
ASD HP	ASD (ADJUSTABLE SPEED DRIVE) HP RATING INDICATED
ASD AMPS OR HP	ASD WITH INTEGRAL DISCONNECTING MEANS AMPS OR HP INDICATED
F 60 A	DISCONNECTING MEANS "B"- CIRCUIT BREAKER "F"- FUSED DISCONNECT "N"- NONFUSED DISCONNECT "S"- MOLDED CASE SWITCH XXA - AMPERE RATING
⊠ 4 FVNR	MOTOR STARTER WITH NEMA SIZE AND TYPE INDICATED
⊠ 4 FVNR	COMBINATION STARTER/DISCONNECT SWITCH IN COMMON ENCLOSURE NEMA SIZE AND TYPE INDICATED
$T	MOTOR RATED SWITCH W/THERMAL OVERLOAD PROTECTION (IF APPLICABLE)
Ⓣ	THERMOSTAT - BY OTHERS (UNLESS OTHERWISE SPECIFIED)
——	CONDUIT - EXPOSED OR SURFACE
– – – –	CONDUIT - EMBEDDED OR BURIED
—●	CONDUIT CHANGES ELEVATION ELEVATIONS NOTED ON PLANS

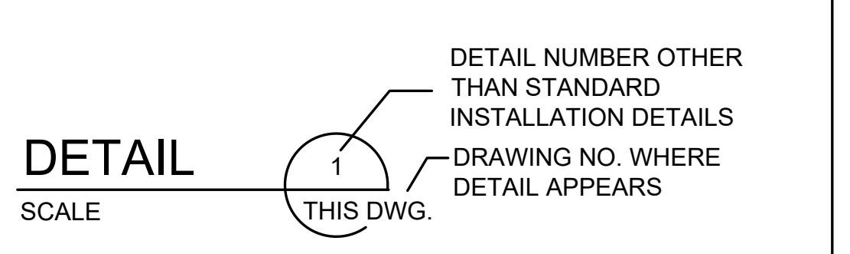
SWITCH LEG AND TRAVELER NUMBERING SCHEMES



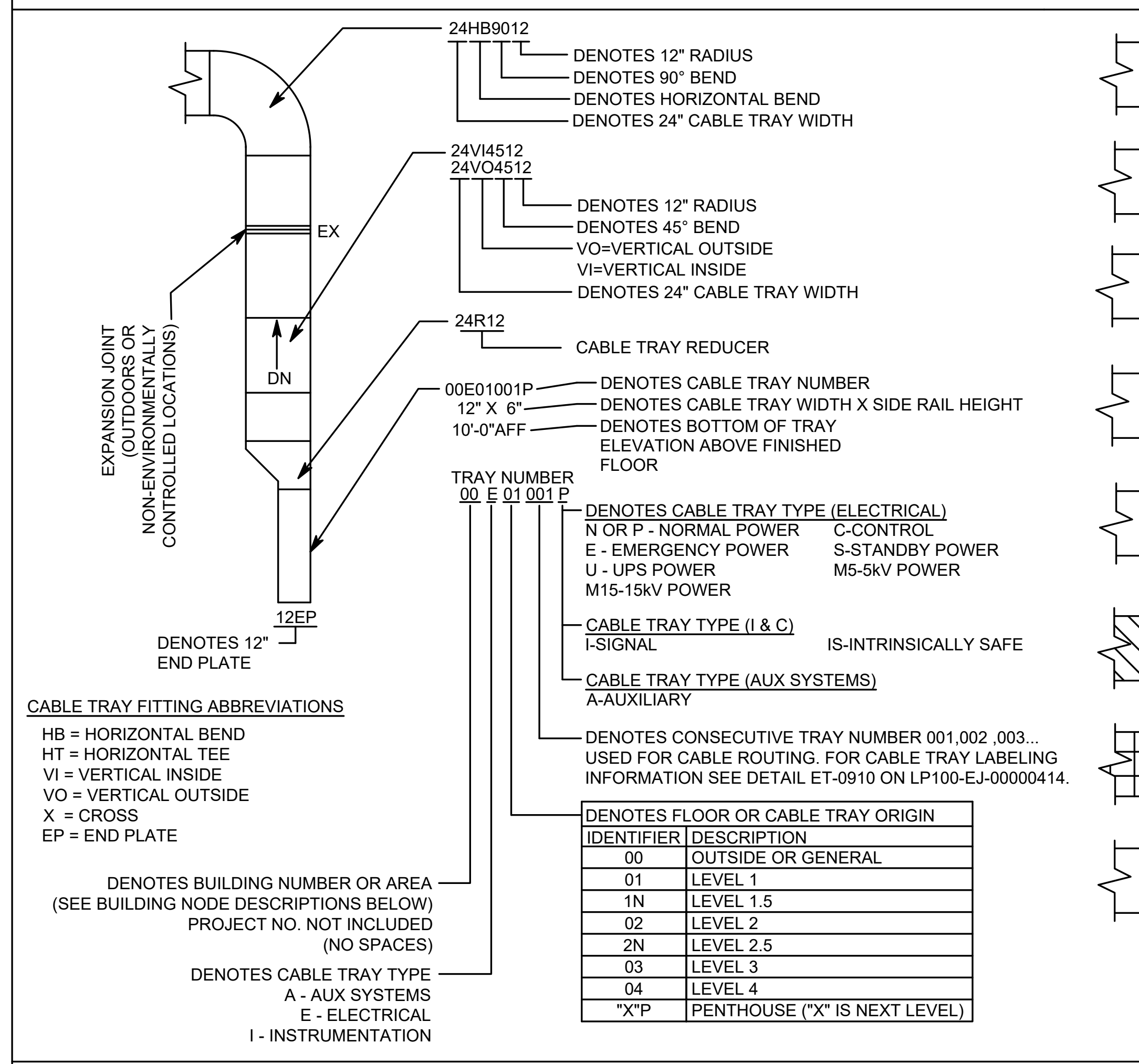
DETAILS, SECTIONS, AND MISCELLANEOUS SYMBOLS


MATCH LINE DWG LP100-EP-00000201
SECTION - A
LP100-EP-00000101
SECTION CUT DETAIL FROM PLAN DWG.
DRAWING WHERE CUT WAS MADE
A
LP100-EJ-00000101
SECTION CUT LETTER
DRAWING WHERE SECTION CUT DETAIL IS LOCATED
DETAIL TYPE SEE TABLE
BLDG. OR AREA (IF APPLICABLE)
901 DETAIL SEQUENCE NUMBER
EG11901
LP131-EJ-00000501
DENOTES LOCATION DRAWING WHERE DETAIL APPEARS

DETAIL TYPE LEGEND	
IDENTIFIER	DESCRIPTION
EA	ABOVEGROUND POWER
EB	BELOW GRADE POWER
EG	GROUNDING
EH	HEAT TRACE
EJ	JUNCTION BOXES
EK	LIGHTING PROTECTION
EL	LIGHTING
EP	POWER
ER	RECEPTACLE/SWITCH
ES	SUPPORTS
ET	TRAY


DETAIL
SCALE
1
THIS DWG.
DETAIL NUMBER OTHER THAN STANDARD INSTALLATION DETAILS
DRAWING NO. WHERE DETAIL APPEARS

CABLE TRAY NUMBERING AND SYMBOLS



DENOTES CABLE TRAY TYPE (ELECTRICAL)
N OR P - NORMAL POWER C-CONTROL
E - EMERGENCY POWER S-STANDBY POWER
U - UPS POWER M5-5kV POWER
M15-15kV POWER

CABLE TRAY TYPE (I & C)
I-SIGNAL IS-INTRINSICALLY SAFE

CABLE TRAY TYPE (AUX SYSTEMS)
A-AUXILIARY

DENOTES CONSECUTIVE TRAY NUMBER 001,002 ,003... USED FOR CABLE ROUTING. FOR CABLE TRAY LABELING INFORMATION SEE DETAIL ET-0910 ON LP100-EJ-00000414.

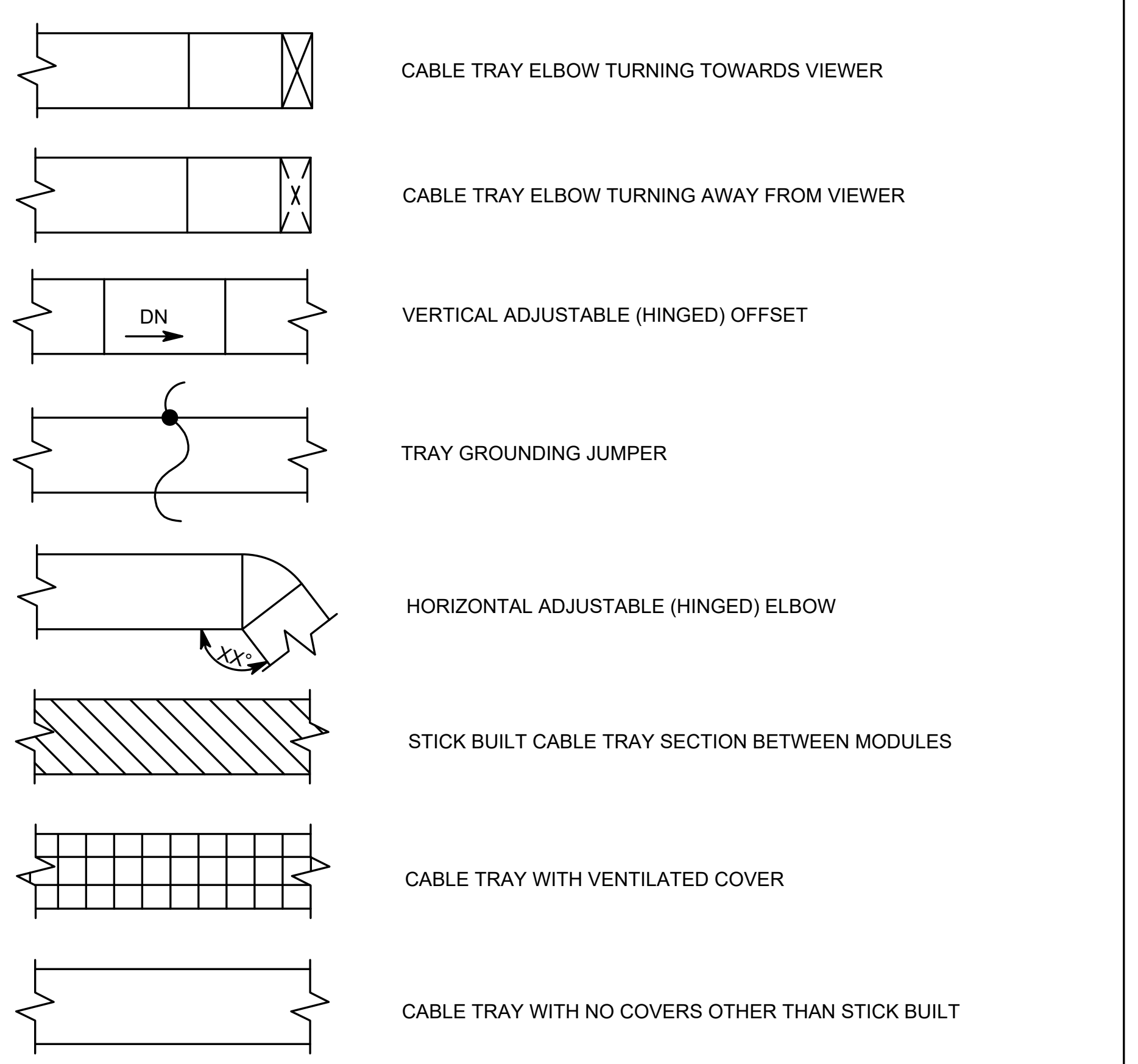
DENOTES FLOOR OR CABLE TRAY ORIGIN	
IDENTIFIER	DESCRIPTION
00	OUTSIDE OR GENERAL
01	LEVEL 1
1N	LEVEL 1.5
02	LEVEL 2
2N	LEVEL 2.5
03	LEVEL 3
04	LEVEL 4
"X"P	PENTHOUSE ("X" IS NEXT LEVEL)

DENOTES BUILDING NUMBER OR AREA (SEE BUILDING NODE DESCRIPTIONS BELOW) PROJECT NO. NOT INCLUDED (NO SPACES)

DENOTES CABLE TRAY TYPE
A - AUX SYSTEMS
E - ELECTRICAL
I - INSTRUMENTATION

CABLE TRAY FITTING ABBREVIATIONS

HB = HORIZONTAL BEND
HT = HORIZONTAL TEE
VI = VERTICAL INSIDE
VO = VERTICAL OUTSIDE
X = CROSS
EP = END PLATE



CABLE TRAY ELBOW TURNING TOWARDS VIEWER

CABLE TRAY ELBOW TURNING AWAY FROM VIEWER

VERTICAL ADJUSTABLE (HINGED) OFFSET

TRAY GROUNDING JUMPER

HORIZONTAL ADJUSTABLE (HINGED) ELBOW

STICK BUILT CABLE TRAY SECTION BETWEEN MODULES

CABLE TRAY WITH VENTILATED COVER

CABLE TRAY WITH NO COVERS OTHER THAN STICK BUILT

ELECTRICAL EQUIPMENT NUMBERING

LP131-MCC/10-10M-01L

PROJECT NUMBER "LP1" NUMBER (TYP. NOT SHOWN FOR CLARITY)

L= LEFT OR R= RIGHT [OPTIONAL] (FACING FRONT OF EQUIPMENT)

SEQUENCE NUMBER 01, 02 ...

COLUMN GRID LINES

NO.	BLDG/NODE DESCRIPTION	NODE ID
00	SITE / GENERAL	GEN
10	PEPTIDE SYS/PUR HEADBLOCK	PHB
11	PEPTIDE SYS/PUR MFG BLDG 1	PP1
11	PEPTIDE SYS A MFG BLDG 1	SA1
11	PEPTIDE SYS B MFG BLDG 1	SB1
12	PEPTIDE SYS/PUR E BLDG 1	PP1
13	PEPTIDE SYS/PUR E BLDG 2	PP1
15	PEPTIDE SYS/PUR MFG BLDG 2	PP2
15	PEPTIDE SYS A MFG BLDG 1	SA2
15	PEPTIDE SYS B MFG BLDG 2	SB2
16	PEPTIDE SYS/PUR E BLDG 1	PP2
17	PEPTIDE SYS/PUR E BLDG 2	PP2
20	OLIGONUCLEOTIDES HEADBLOCK	OLG
21	OLIGONUCLEOTIDES MFG BLDG	OLG
22	OLIGONUCLEOTIDES ELEC BLDG	OLG
30	SMALL MOLECULE HEADBLOCK	SMM
31	SMALL MOLECULE MFG	SMM
32	SMALL MOLECULE ELEC BLDG	SMM
40-49	TANK FARM ELEC BLDGS	VARIOUS
50	CENTRAL UTILITIES (CUB)	CUB
51	MAIN SWITCHGEAR BLDG	CUB
52	COOLING TOWER ELEC ROOM	CUB
53	MODULAR DATA CENTER	STE
55	FIRE & RESCUE CENTER	SMP
56	FIRE PUMP HOUSE BLDG	INT
60	WAREHOUSE BLDG	WHS
61	WAREHOUSE FREEZER BLDG	WHS
65	WASTE RECYCLING CENTER	SMP
70	ADMIN/LAB BLDG	ALB
75	WELCOME CENTER	STE
Y	EXTERIOR PROCESS/SYSTEMS	TFS

FLOOR LEVEL SUB TYPE	
IDENTIFIER	DESCRIPTION
0	GENERAL
B	BASEMENT
M	MEZZANINE
N	INTERSTITIAL
P	PENTHOUSE
R	ROOF
U	UNDER GROUND

FLOOR LEVEL	
IDENTIFIER	DESCRIPTION
0	OUTSIDE OR GENERAL
1	1ST FLOOR
2	2ND FLOOR
3	3RD FLOOR
4	4TH FLOOR
5	5TH FLOOR
6	6TH FLOOR

NOTES:

1. FOR ELECTRICAL EQUIPMENT NUMBERS IDENTIFIED ON P&IDs, SEE FIXED EQUIPMENT NUMBERING SCHEME. (SEE REF. THIS DWG.)
2. FOR ADDITIONAL EQUIPMENT TYPES AND DESCRIPTIONS, SEE ABBREVIATIONS TABLE, THIS DRAWING.

TYPE	EQUIPMENT DESCRIPTION (NOTE 2)
ASD	ADJUSTABLE SPEED DRIVE (SEE NOTE 1)
ATS	AUTOMATIC TRANSFER SWITCH
BC	BATTERY CHARGER WITH BATTERY 125 VDC
BAT	BATTERY CABINET
BD	BUS DUCT
BS	BEST BATTERY SELECTOR (DC POWER)
BTP	BREAKER TEST PANEL
CAP	CAPACITOR BANK
CP	CONTROL PANEL
DCP	DC POWER DISTRIBUTION PANEL
DCTS	DC POWER TIE SWITCH
DS	DISCONNECT SAFETY SWITCH (SEE NOTE 1)
EB	ELECTRICAL BUILDING
ELC	OUTDOOR EMERG. LGT. CONTACTOR PANEL
ELI	EMERGENCY LIGHTING INVERTER
ELP	480Y/277V EMERGENCY LIGHTING PANEL
EHT	ELECTRIC HEAT TRACING
ETX	EMERGENCY TRANSFORMER
EMCC	EMERGENCY MOTOR CONTROL CENTER
EPDP	480V EMERGENCY POWER DIST. PANEL

TYPE	EQUIPMENT DESCRIPTION (NOTE 2)
GEN	STANDBY GENERATOR
HCOM	HEAT TRACE COMMUNICATIONS PANEL
HCP	HEAT TRACE CONTROLLER
HJB	HEAT TRACING JUNCTION BOX
HTX	HEAT TRACE TRANSFORMER
FOP	FIBER OPTIC PATCH PANEL
LC	OUTDOOR LIGHTING CONTACTOR PANEL
LP	480Y/277V LIGHTING PANEL
MBP	MAINTENANCE BYPASS SWITCH
MCC	MOTOR CONTROL CENTER
NGR	NEUTRAL GROUNDING RESISTOR
PDP	480V POWER DISTRIBUTION PANEL
RP	208Y/120V RECEPTACLE PANEL
ROP	REMOTE OPERATING PANEL
SWG	SWITCHGEAR
TX	TRANSFORMER
UDP	UNINTERRUPTIBLE POWER DIST. PANEL
UPP	UNINTERRUPTIBLE POWER SUPPLY PANEL
UPS	UNINTERRUPTIBLE POWER SUPPLY
US	UNIT SUBSTATION
UTX	UNINTERRUPTIBLE PWR. SUPPLY TRANSF.

CONDUIT TAGGING (CONDUIT/EMT/ETC.)

FOR CONDUIT MARKING REQUIREMENTS SEE DWG. LP100-EJ-00000010.

LEGEND: (CONDUIT 2" AND LARGER):

3"C EL. XX'-XX" (BOT. OF CONDUIT ELEV. ABOVE FINISHED SURFACE OR GRADE) IF USED

BOTTOM OFCONDUIT ELEVATION
TRADE SIZE

CABLE TAGGING/IDENTIFICATION

FOR CABLE TAGGING/IDENTIFICATION REQUIREMENTS SEE DWG. LP100-EJ-00000010.

FIXED EQUIPMENT NUMBERING (P&ID)

THIS NUMBERING SCHEME IS FOR REFERENCE ONLY. SEE PIPING & INSTRUMENTATION DIAGRAM LEAD SHEET EQUIPMENT, 5 OF 10, DWG. LP100-DI-00000005, LATEST REVISION FOR FIXED EQUIPMENT TAGGING/IDENTIFICATION.

LP1WW AAA-CCCDEE

SEQUENTIAL ALPHABETIC IDENTIFIER FOR EACH ANCILLARY PIECE OF EQUIPMENT ON DRAWING (DO NOT USE I OR O)
DIGIT TO DIFFERENTIATE BETWEEN SIMILAR EQUIPMENT
LAST THREE DIGITS OF P&ID NUMBER ON WHICH EQUIPMENT IS SHOWN
PREFIX, SEE LP100-DI-00000009
BUILDING NUMBER, IF SHOWN
PROJECT NUMBER (TYP. NOT SHOWN)

EXAMPLES	
TAG/SYMBOL	DESCRIPTION
31-MX-1011A-ASD ASD	MIXER MOTOR 31-MX-101A (ADJ. SPEED DRIVE)
31-MX-1011A-DS	MIXER MOTOR 31-MX-101A (DISCONNECT SWITCH)
31-MX-1011A Ⓜ	MIXER 31-MX-1011A (MOTOR)

EXAMPLE: WELDING RECEPTACLE 31WR WITH FUSED DISCONNECT SWITCH LOCATED ON FIRST FLOOR, *GENERAL* SUB FLOOR TYPE, COL. LINE 12L, SEQ. NO. 01

F
31WR/1012L01 (EQUIP. NO.)
FOR SOURCE SEE CONDUIT/CABLE SCHEDULE OR PANEL SCHEDULE

DRAWING NUMBERING

TYPE	DESCRIPTION
EJ	LEAD SHEETS , FIXTURE SCHEDULE AND STANDARD INSTALLATION DETAILS
EK	ELECTRICAL SINGLE LINE DIAGRAMS, SCHEMATICS, WIRING DIAGRAMS AND AREA CLASSIFICATIONS PLANS, SECTIONS AND DETAILS
EL	ELECTRICAL LIGHTING PLANS, SECTIONS AND DETAILS
EP	ELECTRICAL POWER , ABOVE GROUND, CABLE TRAYS , GROUNDING, UNDERGROUND PLANS, SECTIONS AND DETAILS

EXAMPLE
LP100-EJ-00000002

SEQUENCE NUMBER
DRAWING TYPE
BLDG OR AREA NUMBER (SEE BLDG NODE TABLE, THIS DWG. EQUIPMENT TAGGING SECTION)
PROJECT NUMBER

LIGHTING FIXTURE (LUMINAIRE) NUMBERING SCHEME

AA
BBBBBBBBB - B / B
EL. XX' - XX"

LIGHTING FIXTURE TYPE (SEE FIXTURE SCH. FOR DESC.)
PANELBOARD NO. - CKT.# / SWITCH (IF APPLICABLE)
BOTTOM OF FIXTURE ELEVATION ABOVE FINISHED SURFACE (UNO)

No. PE12300742
SAMUEL BENEFIELD

Covering Rev5, 4Dec2025

Samuel Benefield
Digitally signed by Sam Benefield
Date: 2025.12.04 12:21:09 -05'00'

FLUOR®

EFDEE Engineering PC
PROJECT NO. B1LG
NODE: GEN

REV	DATE	DES	ENG	CHK	APP	PM APP
5	01-DEC-2025	AA	SB	MV	SB	
4	28-OCT-2025	EG	SB	MV	SB	
3	25-AUG-2025	AA	SB	MV	SB	
2	03-MAR-2025	EG	SB	MV	SB	
1	07-MAR-2024	RB	SB	FS	SB	

Lilly

ELECTRICAL
LEGEND AND SYMBOLS
PHYSICAL DRAWINGS AND ABBREVIATIONS

REV	ZONE	PROJECT NUMBER / DESCRIPTION	CHANGE CONTROL	DATE
5	ALL	ISSUED FOR CONSTRUCTION	N/A	01-DEC-2025
4	ALL	ISSUED FOR CONSTRUCTION	N/A	28-OCT-2025
3	ALL	ISSUED FOR CONSTRUCTION	N/A	25-AUG-2025
2	ALL	ISSUED FOR CONSTRUCTION	N/A	03-MAR-2025
1	ALL	ISSUED FOR CONSTRUCTION	N/A	07-MAR-2024

REVISIONS

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SYSTEM TYPE NA	EDIFICE/BUILDING 00	LEVEL/FLOOR 0
SUB DISCIPLINE EJ	LOCATION CODE LP1	LOCATION/GRID
CCI CODE	PROJECT NAME LP1-306-1181	
CRITICAL QUALITY	ENTERPRISE GROUP API	DRAWING INTENTION
DRAWING SCALE NONE	CRITICAL BUSINESS	EQUIPMENT CLASSIFICATION NA
ALTERNATE DOCUMENT ID NA	MERIDIAN REV 5	CRITICAL HSE
VENDOR NA	DOCUMENT ID LP100-EJ-00000002	NON-CRITICAL
ENGINEERING AGENCY DOC ID NA		

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